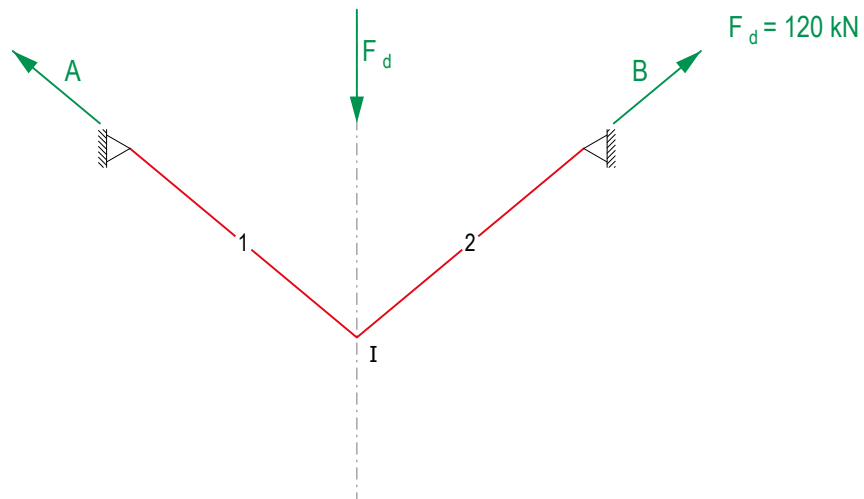


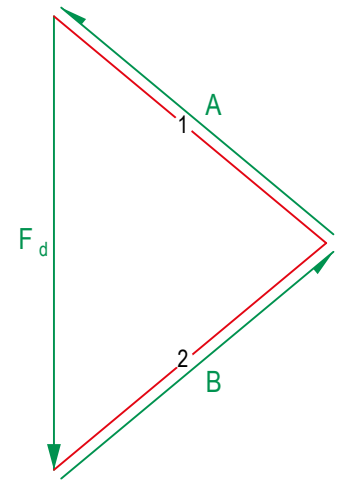
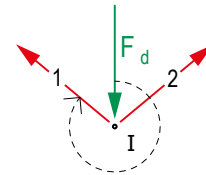
Task 1 Point Load and Uniformly Distributed Load

Design the form of a possible funicular cable for the cases a) and b) and draw the corresponding force diagram. Indicate tension forces with red, compression forces with blue and external forces with green.

a)

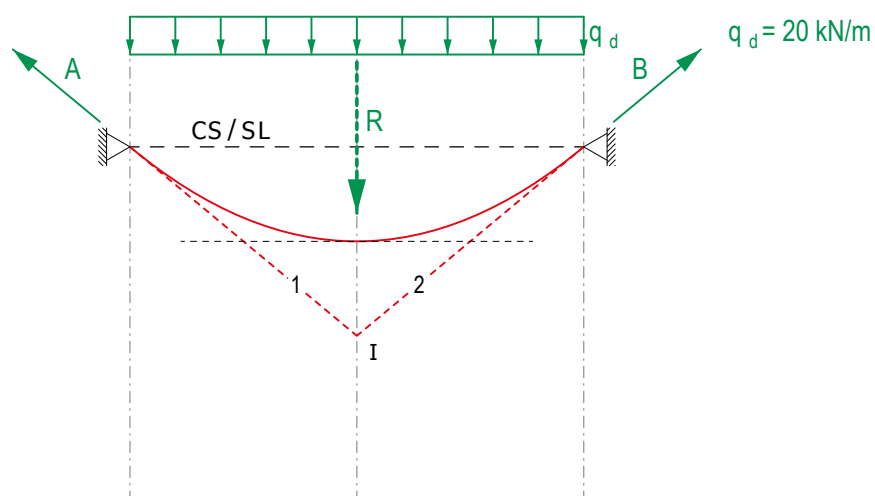


form diagram 1:100

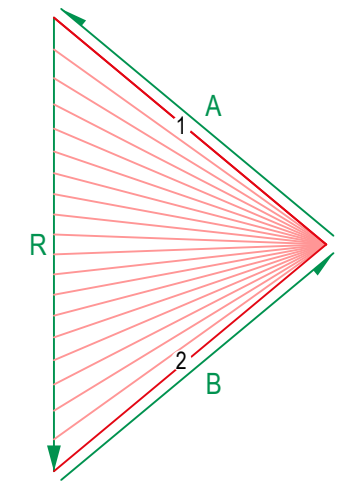
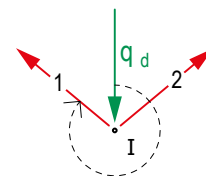


force diagram 1cm $\hat{=}$ 20kN

b)



form diagram 1:100



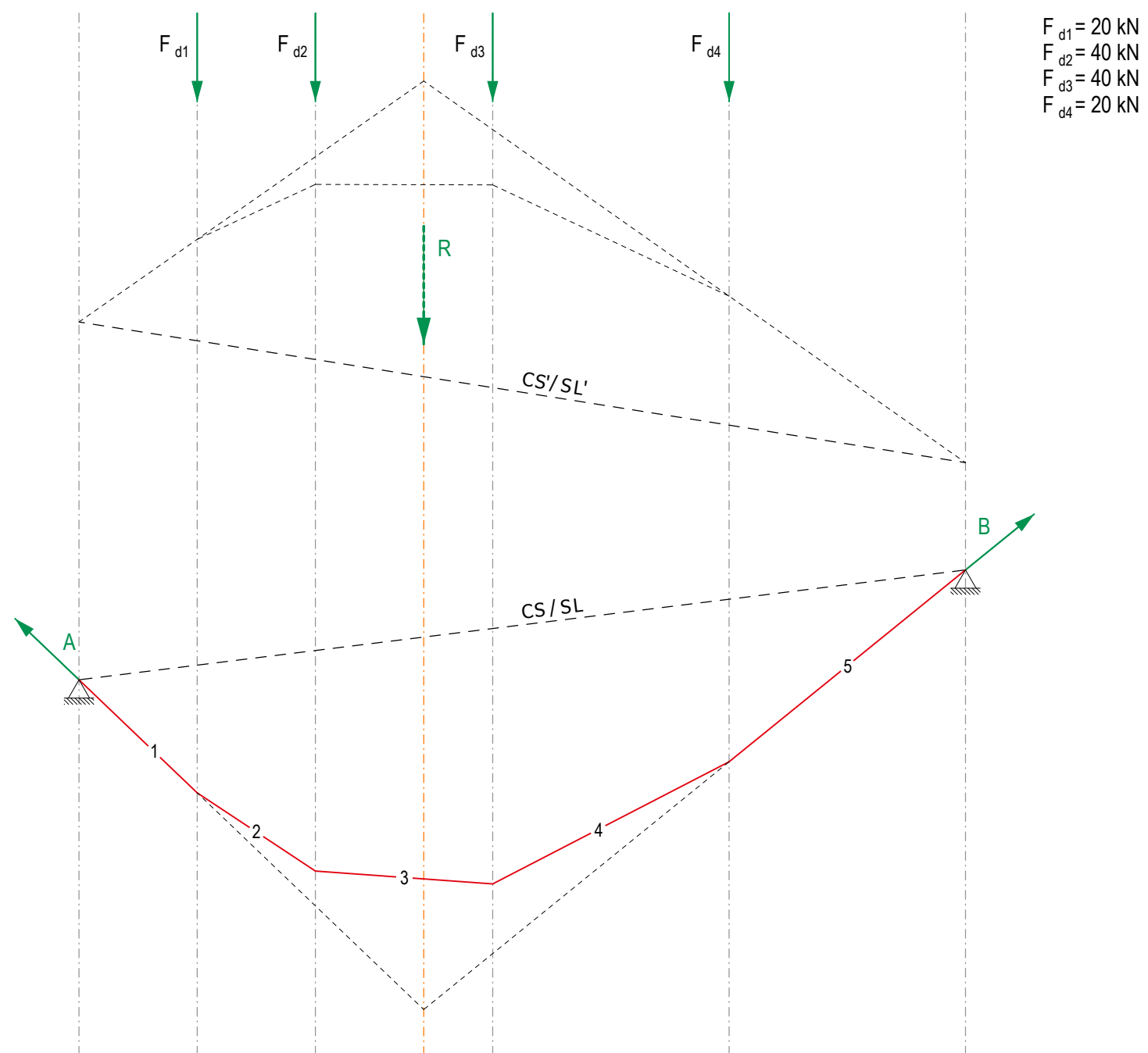
force diagram 1cm $\hat{=}$ 20kN

- c) Is the internal force in case b) constant along the cable? If not, where is the internal force the largest? Draw one additional tangent into the form and force diagram and give reasons for your answer.

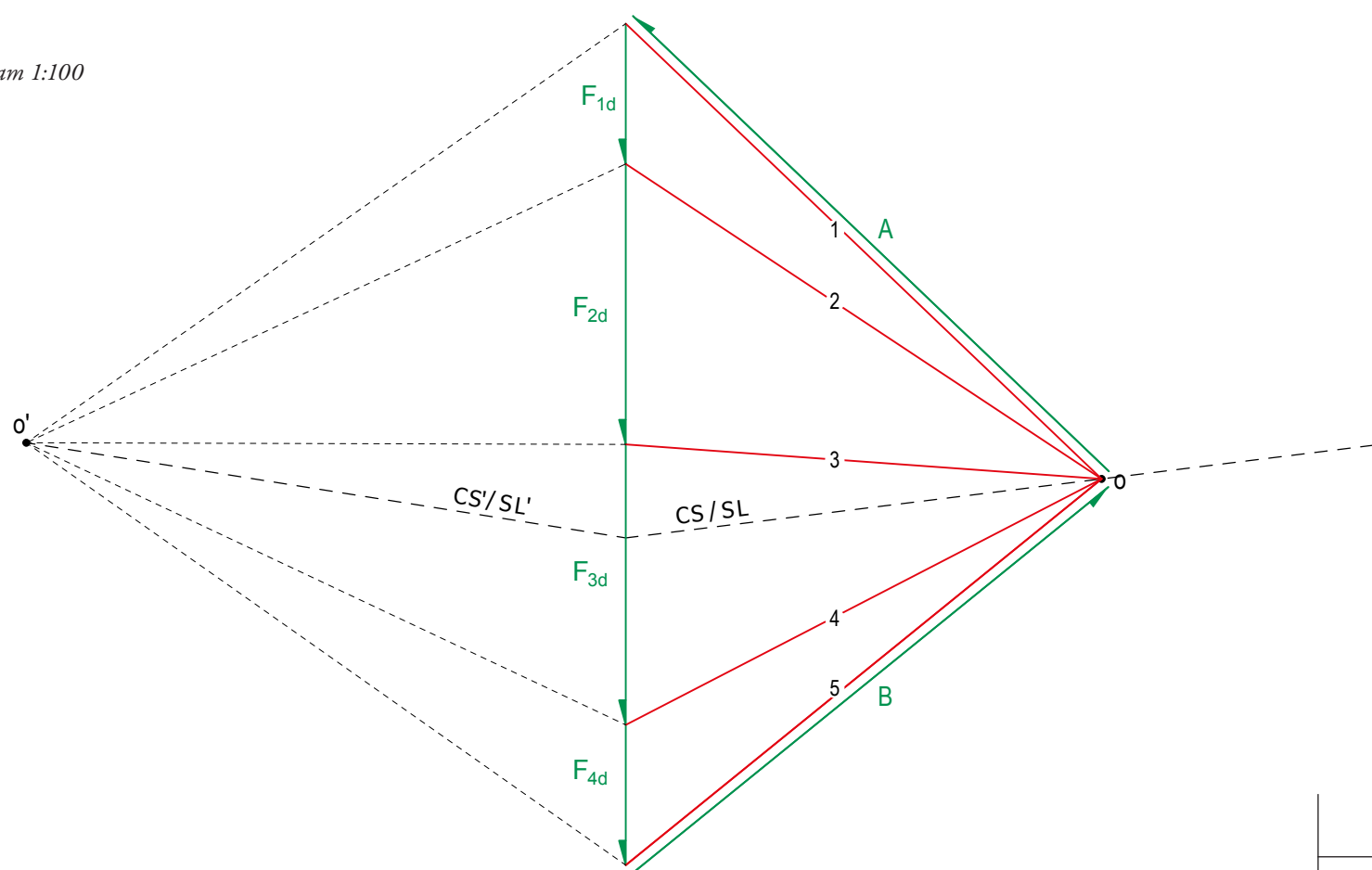
The internal force is not constant along the curve. It is largest at the supports. The largest force corresponds to the outer tangents in the force diagram.

Task 2 Funicular Form

Design the form of a possible funicular cable for the given loading case using the force diagram. Use a trial funicular to find the resultant. Then fix the direction of the reaction forces and find the global equilibrium. Finally find the cable segments. Indicate tension forces with red, compression forces with blue and external forces with green. Furthermore, find the relevant force $N_{d \max}$ and enter it in the table.



form diagram 1:100



$N_{d\max}$ [kN]
94

force diagram $1cm \triangleq 10kN$

Task 3 Designing the Suspended Roof

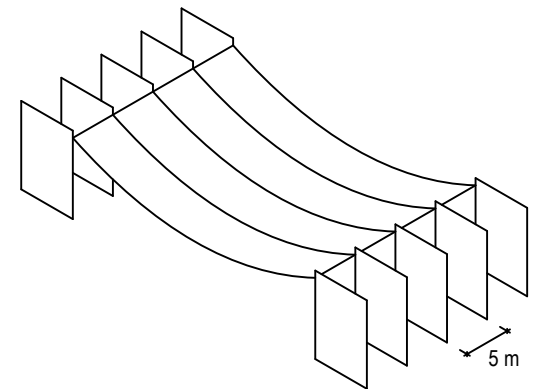
We are analysing the Portuguese National Pavilion by Álvaro Siza. The dead area load is $\bar{g}_k = 5 \text{ [kN/m}^2\text{]}$ and in addition a snow load of $\bar{q}_k = 0.5 \text{ [kN/m}^2\text{]}$ is to be carried.

- First calculate the total constant area load $\bar{s}_d \text{ [kN/m}^2\text{]}$ at design level. ($\bar{s}_d = \bar{g}_d + \bar{q}_d$)
- Calculate the constant line load $s_d \text{ [kN/m]}$ for one of the middle roof elements at a distance of 5m between the supports (structural walls). Thus, first find the tributary area.
- Calculate the magnitude of the resultant from the line load of b).
- Find the form of a funicular tensile roof-structure supported in points A and B such that its lowest point is at 10 meters above ground. Draw the corresponding form and force diagrams. Specify the direction of the support forces. Indicate tension forces with red, compression forces with blue and external forces with green.

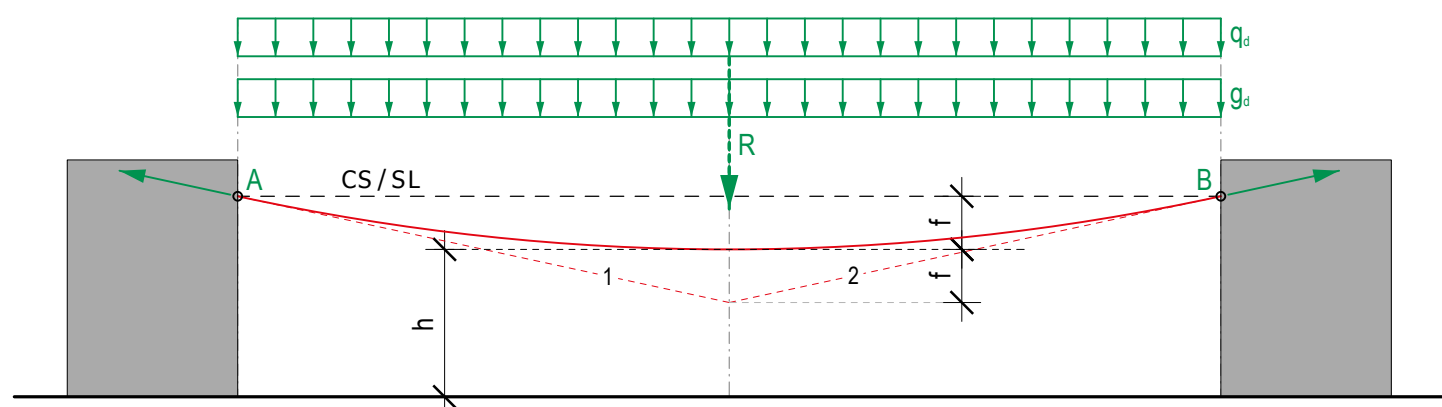
$$\begin{aligned} \bar{s}_d &= \bar{g}_k \cdot \gamma_G + \bar{q}_k \cdot \gamma_Q \\ \bar{s}_d &= 5 \text{ kN/m}^2 \cdot 1.35 + 0.5 \text{ kN/m}^2 \cdot 1.5 = \mathbf{7.5 \text{ kN/m}^2} \end{aligned}$$

$$\begin{aligned} s_d &= \bar{s}_d \cdot b \\ s_d &= 7.5 \text{ kN/m}^2 \cdot 5 \text{ m} = \mathbf{37.5 \text{ kN/m}} \end{aligned}$$

$$\begin{aligned} R &= s_d \cdot l \\ R &= 37.5 \text{ kN/m} \cdot 65 \text{ m} = \mathbf{2437.5 \text{ kN}} \end{aligned}$$



d)



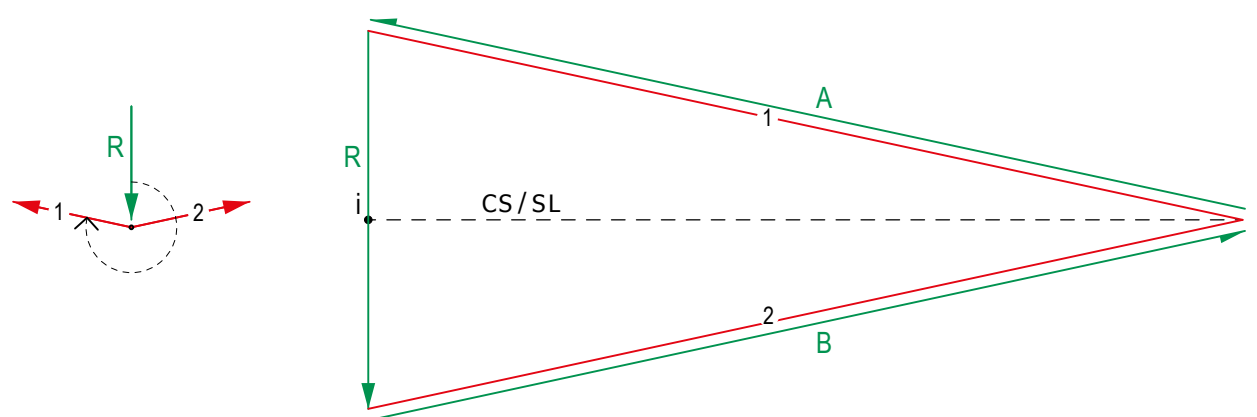
form diagram 1:500

internal forces:

$$N_{d\max} = N_1 = N_2 = 5915 \text{ kN}$$

Reaction forces:

$$A = B = 5915 \text{ kN}$$



force diagram 1cm $\hat{=}$ 500kN

- How do the internal forces in the cable change if the loading is doubled but the form remains the same?

As the loading is doubled, the whole force diagram will be scaled by the factor 2, therefore the internal forces in the cable will double as well.

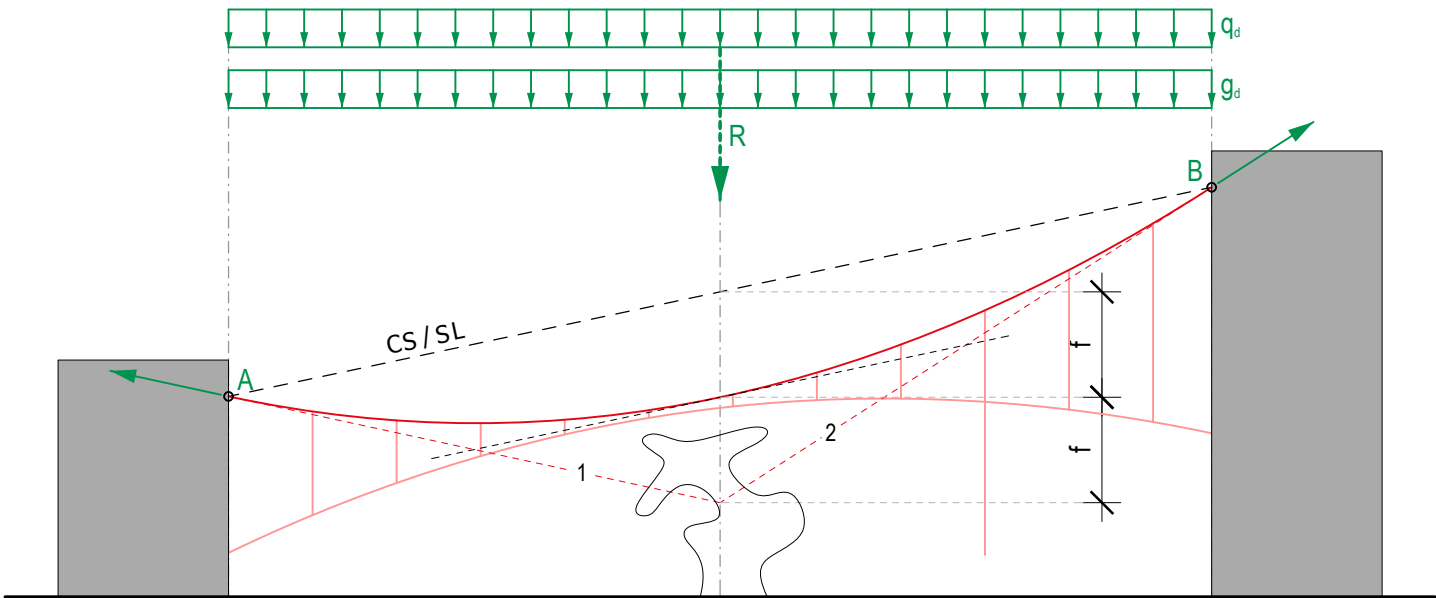
$$N_{d\max} = N_{d1} = N_{d2} = 11'830 \text{ kN}$$

Creative Task

Designing an Asymmetric Suspended Roof

The sculpture of a well-known artist is supposed to be displayed in the pavilion. But the room height of the first design is not sufficient for the sculpture to fit. The loading case is equal to the one in task 3.

- a) Choose a new height for the right support wall and design the new form of the funicular tensile roof-structure. Draw the corresponding form and force diagrams. Find the magnitude and direction of the reaction forces. Indicate tension forces with red, compression forces with blue and external forces with green.
- b) The suspended roof is built from a cable made out of steel S235. Calculate the diameter of the cable considering its relevant internal force.
- c) Sketch a possible stiffening scheme for the roof into the form diagram.

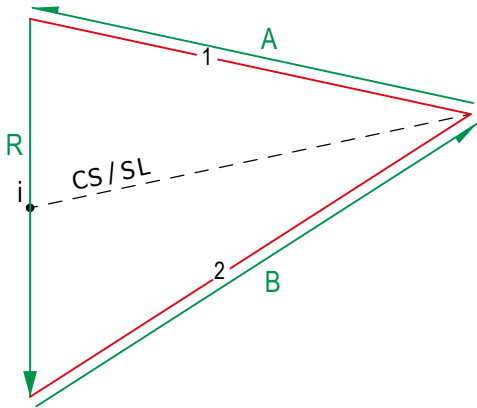
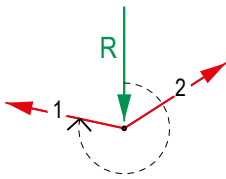


form diagram 1:500

A [kN]	B [kN]
3000	3500

internal forces:
 $N_1 = 3000 \text{ kN}$
 $N_2 = N_{\max} = 3500 \text{ kN}$

Reaction forces
A = 3500 kN
B = 3500 kN



force diagram 1cm ≙ 500kN

b)

$$N_{d,\max} = 3'500'000 \text{ N}$$
$$f_{td} = 223.8 \text{ N/mm}^2$$
$$A_{\text{req}} = \frac{N_d}{f_{td}} = 15'638.96 \text{ mm}^2$$
$$D = \sqrt{4 \cdot A / \pi} = 142 \text{ mm}$$